

SAFETY DATA SHEET

(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 453/2010

Revision Date 01-Apr-2016

WAI2 - EGHS - EUROPEAN

Revision Number 3

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name Nitrate 1 (as N)-LR
Product No AC4004-AMP
Pure substance/mixture Mixture

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Orion Inc. (Part of Thermo Fisher Scientific, Inc.)
Water Analysis Instruments
22 Alpha Road
Chelmsford, MA 01824, USA
1-978-232-6000

E-mail address wai.techservbev@thermofisher.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture**

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Serious eye damage/eye irritation	Category 2 - (H319)
Acute aquatic toxicity	Category 3 - (H411)

2.2. Label elements**Signal Word**

Warning

Hazard Statements

H319 - Causes serious eye irritation

H411 - Toxic to aquatic life with long lasting effects

Precautionary Statements

P202 - Do not handle until all safety precautions have been read and understood

2.3. Other hazards

May be harmful if swallowed

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.1. Substances**

Component	Chemical Formula	EC-No.	CAS-No	Weight %	CLP Classification - Regulation (EC) No 1272/2008	REACH Reg. No
Water	No information available	EEC No. 231-791-2	7732-18-5	80 - 90%		No information available
Ethylene Glycol	No information available	EEC No. 203-473-3	107-21-1	10 - 20%	Acute Tox. 4 (H302)	No information available
2-Propanol	No information available	EEC No. 200-661-7	67-63-0	0 - 10%	Eye Irrit. 2 (H319) STOT SE 3 (H336) Flam. Liq. 2 (H225)	No information available
Sulfanilic Acid	No information available	EEC No. 208-208-5	515-74-2	0 - 10%		No information available
2,5 Dihydroxybenzoic Acid	No information available	EEC No. 207-718-5	490-79-9	0 - 10%		No information available
Disodium 4,5-dihydroxynaphthalene-2,7-disulphonate	No information available	-	5808-22-0	0 - 10%		No information available
Cyclohexanediamine Tetraacetic Acid, Disodium Magnesium Salt	No information available	EEC No. 264-195-6	63451-33-2	0 - 10%		No information available
Proprietary Ingredients	No information available	-	999-99-9	0 - 10%		No information available

Note *The exact percentage (concentration) of composition has been withheld as a trade secret

Full text of H- and EUH-phrases: see section 16

SECTION 4: FIRST AID MEASURES**4.1. Description of first aid measures**

General Advice	Use first aid treatment according to the nature of the injury. For further assistance, contact your local Poison Control Center. Show this safety data sheet to the doctor in attendance.
Eye Contact	In case of eye contact, remove contact lens and rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If symptoms persist, call a physician.
Inhalation	Move to fresh air. If breathing is difficult, give oxygen. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting. Call a physician or Poison Control Center immediately.
Protection of First-aiders	Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms/effects No information available

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES**5.1. Extinguishing media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES**6.1. Personal precautions, protective equipment and emergency procedures**

Personal Precautions Use personal protective equipment. Evacuate personnel to safe areas.

6.2. Environmental precautions

Environmental Precautions Avoid release to the environment.

6.3. Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling

To avoid risks to human health and the environment, comply with the instructions for use. Wear personal protective equipment. Avoid breathing dust/fume/gas/mist/vapors/spray. Ensure adequate ventilation, especially in confined areas.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Keep away from direct sunlight.

7.3. Specific end use(s)

Specific Use(s)

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Component	European Union	The United Kingdom	France	Spain	Germany
Ethylene Glycol 107-21-1	TWA: 20 ppm 8 hr TWA: 52 mg/m ³ 8 hr STEL: 40 ppm 15 min STEL: 104 mg/m ³ 15 min Possibility of significant uptake through the skin	STEL: 40 ppm 15 min STEL: 104 mg/m ³ 15 min STEL: 30 mg/m ³ 15 min TWA: 10 mg/m ³ 8 hr TWA: 20 ppm 8 hr TWA: 52 mg/m ³ 8 hr Skin	TWA / VME: 20 ppm (8 heures). indicative limit TWA / VME: 52 mg/m ³ (8 heures). indicative limit STEL / VLCT: 40 ppm. indicative limit STEL / VLCT: 104 mg/m ³ . indicative limit Peau	STEL / VLA-EC: 40 ppm (15 minutos). STEL / VLA-EC: 104 mg/m ³ (15 minutos). TWA / VLA-ED: 20 ppm (8 horas) TWA / VLA-ED: 52 mg/m ³ (8 horas) Piel	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2 TWA: 26 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time TWA: 26 mg/m ³ (8 Stunden). MAK can occur as vapor and aerosol at the same time Höhepunkt: 20 ppm Höhepunkt: 52 mg/m ³ Haut
2-Propanol	-	STEL: 500 ppm 15 min	STEL / VLCT: 400	STEL / VLA-EC: 400	TWA: 200 ppm (8

67-63-0		STEL: 1250 mg/m ³ 15 min TWA: 400 ppm 8 hr TWA: 999 mg/m ³ 8 hr	ppm. STEL / VLCT: 980 mg/m ³ .	ppm (15 minutos). STEL / VLA-EC: 1000 mg/m ³ (15 minutos). TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 500 mg/m ³ (8 horas)	Stunden). AGW - exposure factor 2 TWA: 500 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 200 ppm (8 Stunden). MAK TWA: 500 mg/m ³ (8 Stunden). MAK Höhepunkt: 400 ppm Höhepunkt: 1000 mg/m ³
Component	Italy	Portugal	The Netherlands	Finland	Denmark
Ethylene Glycol 107-21-1	TWA: 20 ppm 8 ore. Media Ponderata nel Tempo TWA: 52 mg/m ³ 8 ore. Media Ponderata nel Tempo STEL: 40 ppm 15 minuti. Breve termine STEL: 104 mg/m ³ 15 minuti. Breve termine Pelle	STEL: 40 ppm 15 minutos STEL: 104 mg/m ³ 15 minutos Ceiling: 100 mg/m ³ TWA: 20 ppm 8 horas TWA: 52 mg/m ³ 8 horas Pele	huid STEL: 104 mg/m ³ 15 minuten TWA: 52 mg/m ³ 8 uren TWA: 10 mg/m ³ 8 uren	TWA: 20 ppm 8 tunteina TWA: 50 mg/m ³ 8 tunteina STEL: 40 ppm 15 minuutteina STEL: 100 mg/m ³ 15 minuutteina Iho	TWA: 10 ppm 8 timer TWA: 26 mg/m ³ 8 timer TWA: 10 mg/m ³ 8 timer Hud
2-Propanol 67-63-0	-	STEL: 400 ppm 15 minutos TWA: 200 ppm 8 horas	-	TWA: 200 ppm 8 tunteina TWA: 500 mg/m ³ 8 tunteina STEL: 250 ppm 15 minuutteina STEL: 620 mg/m ³ 15 minuutteina	TWA: 200 ppm 8 timer TWA: 490 mg/m ³ 8 timer
Component	Austria	Switzerland	Poland	Norway	Ireland
Ethylene Glycol 107-21-1	Haut MAK-KZW: 20 ppm 15 Minuten MAK-KZW: 52 mg/m ³ 15 Minuten MAK-TMW: 10 ppm 8 Stunden MAK-TMW: 26 mg/m ³ 8 Stunden	Haut/Peau STEL: 20 ppm 15 Minuten STEL: 52 mg/m ³ 15 Minuten TWA: 10 ppm 8 Stunden TWA: 26 mg/m ³ 8 Stunden	STEL: 50 mg/m ³ 15 minutach TWA: 15 mg/m ³ 8 godzinach	TWA: 20 mg/m ³ 8 timer TWA: 52 ppm 8 timer TWA: 52 mg/m ³ 8 timer STEL: 52 mg/m ³ 15 minutter. Norm is based on the sum calculation for the total gas and particulate form of the substance dust STEL: 20 ppm 15 minutter. Norm is based on the sum calculation for the total gas and particulate form of the substance Hud	TWA: 10 mg/m ³ 8 hr. particulate TWA: 20 ppm 8 hr. vapour TWA: 52 mg/m ³ 8 hr. vapour STEL: 40 ppm 15 min STEL: 104 mg/m ³ 15 min Skin
2-Propanol 67-63-0	MAK-KZW: 800 ppm 15 Minuten MAK-KZW: 2000 mg/m ³ 15 Minuten MAK-TMW: 200 ppm 8 Stunden MAK-TMW: 500 mg/m ³ 8 Stunden	STEL: 400 ppm 15 Minuten STEL: 1000 mg/m ³ 15 Minuten TWA: 200 ppm 8 Stunden TWA: 500 mg/m ³ 8 Stunden	STEL: 1200 mg/m ³ 15 minutach TWA: 900 mg/m ³ 8 godzinach	TWA: 100 ppm 8 timer TWA: 245 mg/m ³ 8 timer STEL: 100 ppm 15 minutter. STEL: 245 mg/m ³ 15 minutter.	TWA: 200 ppm 8 hr. STEL: 400 ppm 15 min Skin

Derived No Effect Level (DNEL) No information available

Predicted No Effect Concentration (PNEC) No information available

8.2. Exposure controls

Engineering Measures Showers

Eyewash stations
Ventilation systems

Personal protective equipment

Eye/face Protection	Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:. Goggles.
Skin and body protection	Wear protective gloves/clothing.
Respiratory Protection	No protective equipment is needed under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Brownish yellow to gold
Odor	Odorless
Odor Threshold	No information available
PH Range	2-5

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	125 °C / 257 °F	
Flash Point (High in °C)	No information available	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature		
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact None
Sensitivity to Static Discharge None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on toxicological effects****Acute Toxicity****Product Information**

Product does not present an acute toxicity hazard based on known or supplied information.

Inhalation	No information available
Eye Contact	No information available
Skin Contact	No information available
Ingestion	No information available

Unknown Acute Toxicity 3.9 % of the mixture consists of ingredient(s) of unknown toxicity.

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral)	4,594.00 mg/kg
ATEmix (dermal)	59,436.00 mg/kg
ATEmix (inhalation-dust/mist)	2,420.00 mg/L

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	LD50 > 90 mL/kg (Rat)		
Ethylene Glycol	LD50 = 4700 mg/kg (Rat)	LD50 = 9530 µL/kg (Rabbit) LD50 = 10600 mg/kg (Rat)	
2-Propanol	LD50 = 1870 mg/kg (Rat)	LD50 = 4059 mg/kg (Rabbit)	LC50 = 72600 mg/m ³ (Rat) 4 h
Sulfanilic Acid	LD50 = 12300 mg/kg (Rat)		

Skin Corrosion/Irritation No information available

Serious eye damage/eye irritation No information available

Sensitization No information available

Mutagenic Effects No information available

Carcinogenic effects No information available

Reproductive Effects No information available

STOT - single exposure No information available

STOT - repeated exposure No information available

Aspiration hazard No information available

SECTION 12. ECOLOGICAL INFORMATION

12.1. Toxicity

3.9% of the mixture consists of component(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Ethylene Glycol	EC50: 6500 - 13000 mg/L, 96h (Pseudokirchneriella subcapitata)	LC50: = 16000 mg/L, 96h static (Poecilia reticulata) LC50: 40000 - 60000 mg/L, 96h static (Pimephales promelas) LC50: = 27540 mg/L, 96h static (Lepomis macrochirus) LC50: 14 - 18 mL/L, 96h static (Oncorhynchus mykiss) LC50: = 41000 mg/L, 96h (Oncorhynchus mykiss) LC50: = 40761 mg/L, 96h static (Oncorhynchus mykiss)	EC50: = 46300 mg/L, 48h (Daphnia magna)
2-Propanol	EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus) EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)	LC50: > 1400000 µg/L, 96h (Lepomis macrochirus) LC50: = 11130 mg/L, 96h static (Pimephales promelas) LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)	EC50: = 13299 mg/L, 48h (Daphnia magna)
Sulfanilic Acid	EC50: = 91 mg/L, 72h (Desmodesmus subspicatus)	LC50: = 100.4 mg/L, 96h static (Pimephales promelas)	EC50: = 85.66 mg/L, 48h (Daphnia magna)

12.2. Persistence and degradability

No information available

12.3. Bioaccumulative potential

No information available

Component	log Pow
Ethylene Glycol	-1.93
2-Propanol	0.05

12.4. Mobility in soil

No information available

Mobility

12.5. Results of PBT and vPvB assessment

No information available

12.6. Other adverse effects

No information available

Endocrine Disruptor Information

No information available

Component	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
2,5 Dihydroxybenzoic Acid	Group III Chemical	-	-

SECTION 13. DISPOSAL CONSIDERATIONS**13.1. Waste treatment methods**

Waste from Residues / Unused Products Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Improper disposal or reuse of this container may be dangerous and illegal.

SECTION 14: TRANSPORT INFORMATION**IMDG/IMO**

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Marine Pollutant	Not Applicable
14.6 Special Provisions	None
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	No information available

ICAO

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

IATA

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

SECTION 15: REGULATORY INFORMATION**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****European Union**

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

International Inventories

USINV	Complies
CANINV	Does not Comply
EINECS/ELINCS	Does not Comply
ENCS	Does not Comply
IECSC	Does not Comply
KECL	Does not Comply
PICCS	Does not Comply
AICS	Does not Comply

USINVI - TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
CANINVI/ DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
ENCS - Japanese Existing and New Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
AICS - Australian Inventory of Chemical Substances

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION

Key or legend to abbreviations and acronyms used in the safety data sheet

Full text of H-Statements referred to under section 3

H302 - Harmful if swallowed
H319 - Causes serious eye irritation
H336 - May cause drowsiness or dizziness
H225 - Highly flammable liquid and vapor

Legend - SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

TWA	TWA (time-weighted average)	STEL	STEL (Short Term Exposure Limit)
Ceiling	Maximum limit value	*	Skin designation

Prepared By Environmental, Health and Safety

Prepared For Thermo Fisher Scientific Inc.

Issue Date No information available

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Reason for revision SDS sections updated.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

Disclaimer

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End of Safety Data Sheet